

Maria DeCaro

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EDUCATION

Cornell Tech (Cornell University), New York, NY
Jacobs Technion-Cornell Dual Master of Science Degrees
Information Science and Applied Information Systems

August 2024 - May 2026

Syracuse University, Syracuse, NY
Bachelor of Science in Information Management & Technology
Concentrations: Data Analytics and Project Management
Renée Crown University Honors Program

August 2022 - May 2024

TECHNICAL SKILLS

Coding Languages: Python, SQL, R, HTML & CSS
Libraries & Tools: pandas, NumPy, scikit-learn, LightGBM, Matplotlib, Seaborn, Plotly, NetworkX, BeautifulSoup, Jupyter Notebooks, Microsoft Excel (Office Specialist certified), Git, BERTopic, OpenAI API
Techniques: Machine learning (classification, clustering, regression, ensembles, deep learning, logistic regression, random forest, gradient boosting), NLP, sentiment analysis, topic modeling, anomaly detection, feature engineering, A/B testing, hypothesis testing, statistical modeling, visualization, web scraping

WORK EXPERIENCE

Federal Bureau of Investigation: FBI Honors Intern (Albany, NY, Baltimore, MD & New York, NY) June 2024 - Present

- **Albany, NY (Summer 2024):** Developed a complex data analytics tool to simplify identifying relationships and degrees of connection within a specific domain, using Python and key data analytics modules. Created network visualizations with Matplotlib and NetworkX to uncover patterns and common characteristics. Additionally, leveraged Excel for data organization, trend analysis, and reporting, improving the efficiency of investigative workflows.
- **Baltimore, MD (Summer 2025):** Applied data science to investigations transforming large JSON returns into structured datasets using Python (pandas, regex), optimizing scripts for faster processing. Built structured databases merging NASA and FBI evidence to enable pattern analysis and trial preparation. Supported Cyber and CART teams by researching ransomware targets on the dark web and archiving digital evidence to preserve forensic integrity.
- **New York, NY (Academic Year 2024 – 2025, Ongoing):** Split time between two teams:
 - Computer Analysis Response Team (CART): Learn critical forensic software, assist with various technical requests, and support digital evidence processing.
 - Data Analyst Squad: Develop Python programs, including web scraping with BeautifulSoup and automation tools, to assist agents with investigations. Work with large datasets, improving data intake using Python and internal systems.

RESEARCH EXPERIENCE

Cornell Tech STech Lab: AI and Petition Analytics Research January 2025 - Present

- Collaborated with a research team to analyze 700K+ Change.org petitions, evaluating how AI tools shaped user engagement and petition success by comparing adoption patterns, petition scope, and changes over time.
- Built classification pipelines with OpenAI agents and clustering methods (e.g., BERTopic, semantic similarity, K-means) to identify high-engagement topics and categorize petitions, clarifying how issue type shaped verifiable content and impact.

RELEVANT PROJECTS

Cornell Tech Applied Machine Learning: Demand Estimation and Pricing Optimization Project Fall 2025

- Built and evaluated predictive models to estimate customer purchase probability, using cross-validation and feature engineering to improve performance and achieve ROC-AUC up to 0.99, informing data-driven pricing decisions.
- Developed a dynamic, competitor-aware pricing framework based on expected revenue, enabling optimized price selection and adaptive responses to market conditions to maximize overall profit in a simulated environment.

Cornell Tech Data Science in the Wild: Yelp Review Classification Project Spring 2025

- Collaborated in a team of four to analyze nearly 7 million Yelp reviews across 40+ columns, leading data cleaning, preprocessing, and structured organization of review, user, and business metadata to support machine learning workflows.
- Designed and implemented the top-performing models using OpenAI's API for zero-shot, few-shot, and fine-tuned classification (achieving 75.5% accuracy), and supported tuning of a LightGBM model as the top non-AI baseline.

Cornell Tech Trust & Safety: Labeling Giveaway Scams on Bluesky Spring 2025

- Collaborated in a team of three using pair programming to build a rule-based labeler and design a multi-tiered content moderation policy, integrating platform data and safety principles to detect and mitigate fraudulent giveaway scams.
- Led bot detection, replacing Isolation Forest with threshold-based regression on account data, achieving 89.5% accuracy.